数学分析 B2 期中复习.

第8章 空间解析几何

1. $\vec{l_1} = (-1,1,-2)$ $\quad M_1(1,-1,-6)$

$\vec{l_2} = (3,-1,1)$ $\quad M_2(-2,1,1)$

$\vec{l} = (a,b,c)$. $\quad M_0(22,0,2)$

L_1与l共面 $\Longleftrightarrow 0=(\vec{l_1}\times\vec{l})\cdot\overrightarrow{M_1M_0} = \begin{vmatrix} 22-1 & 0-(-1) & 2-(-6) \\ -1 & 1 & -2 \\ a & b & c\end{vmatrix}$

$= 21(c+2b)+(-2a+c)+8(-b-a)$

即 $5a-17b-11c=0$

L_2与l共面 $\Longleftrightarrow 0=(\vec{l_2}\times\vec{l})\cdot\overrightarrow{M_2M_0} = \begin{vmatrix} 22-(-2) & 0-1 & 2-1 \\ 3 & -1 & 1 \\ a & b & c\end{vmatrix}$

$= 24(-c-b)-(a-3c)+(3b+a)$

即 $b+c=0$

取$c=-5 \Rightarrow a=6$, $b=5$

$\vec{l}=(6,5,-5)$.

$$\begin{cases} \frac{x-22}{6}=\frac{y}{5}=\frac{z-2}{-5} \\ \frac{x-1}{-1}=y+1=\frac{z+6}{-2} \end{cases} \Rightarrow \begin{cases} x=10 \\ y=-10 \\ z=12 \end{cases}$$

方向 $\vec{l}=(6,5,-5)$, 交点 $(10,-10,12)$

2.

平面法向量 $\vec{n}=(1,-1,2)$

直线 $\vec{l}=(1,1,-1)$

$\vec{l_{\parallel}}=\vec{n}\times(\vec{n}\times\vec{l})=(-8,-4,2)=2(-4,-2,1)$

取 $\vec{l_0}=(4,2,-1)$.

$$\begin{cases} x-1=y \\ -y=z-1 \\ x-y+2z-1=0 \end{cases} \Rightarrow \begin{cases} x=2 \\ y=1 \\ z=0 \end{cases}$$

$l_0: \frac{x-2}{4}=\frac{y-1}{2}=\frac{z}{-1}$

绕Oy轴旋转 $x^2+z^2=(2y)^2+\left(\frac{-y+1}{2}\right)^2$

$\Rightarrow 4x^2+4z^2=16y^2+y^2-2y+1$

即, $4x^2-17y^2+4z^2+2y-1=0$

3. $\vec{l_0}=(0,\frac{1}{2},1)$

交点(垂足) $M_0(1,t,2t+\frac{1}{2})$ $\quad \overrightarrow{MM_0}=(1,t+1,2t-\frac{1}{2})$

$\overrightarrow{MM_0}\perp\vec{l_0} \Longleftrightarrow 0=\overrightarrow{MM_0}\cdot\vec{l_0}=0+\frac{1}{2}(t+1)+(2t-\frac{1}{2})$

$\Rightarrow t=0$

$\vec{l}\parallel\overrightarrow{MM_0}=(1,1,-\frac{1}{2})$ $\quad$ 取 $\vec{l}=(2,2,-1)$.

$l: \frac{x}{2}=\frac{y+1}{2}=\frac{z-1}{-1}$

平面 $\vec{n}=(a,b,c)$. $\quad$ 平面$y=0$法向量 $\vec{n_1}=(0,1,0)$

$\vec{n}\perp\vec{l} \Longleftrightarrow 0=\vec{n}\cdot\vec{l}=2a+2b-c$

$\vec{n}\perp\vec{n_1} \Longleftrightarrow 0=\vec{n}\cdot\vec{n_1}=b$

取$a=1$, $\Rightarrow b=0$, $c=2$.

$\vec{n}=(1,0,2)$.

π: $(x-0)+0+2(y-1)=0$ 即 $x+2y-2=0$.

4. (1). $\overrightarrow{AB}=(1,2,1)$, $\overrightarrow{AC}=(2,2,0)$, $\overrightarrow{AD}=(3,1,0)$

$V=\frac{1}{6}|(\overrightarrow{AB}\times\overrightarrow{AC})\cdot\overrightarrow{AD}|=\frac{2}{3}$

(2) $\overrightarrow{BC}=(1,0,-1)$. $\quad \overrightarrow{BD}=(2,-1,-1)$

$\vec{n}\parallel\overrightarrow{BC}\times\overrightarrow{BD}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & -1 \\ 2 & -1 & -1 \end{vmatrix}=(-1,-1,-1)$

取 $\vec{n}=(1,1,1)$.

l: $x-1=y=z$

(3) 绕z轴旋转 $x^2+y^2=(z+1)^2+z^2=2z^2+2z+1$

即 $x^2+y^2-2z^2-2z-1=0$.

5. 平面法向量 $\vec{n}=(a,b,c)$.

$\vec{l}=\vec{n_1}\times\vec{n_2}=\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & -1 \\ 3 & -2 & -2 \end{vmatrix}=(-4,1,-7)$

$\vec{n_0}=(3,2,3)$

$\vec{n}\perp\vec{n_0} \Longleftrightarrow 0=\vec{n}\cdot\vec{n_0}=3a+2b+3c$

$\vec{n}\perp\vec{l} \Longleftrightarrow 0=\vec{n}\cdot\vec{l}=-4a+b-7c$

取$c=1$. $\Rightarrow a=-\frac{17}{11}$, $b=\frac{9}{11}$

取 $\vec{n}=(17,-9,-11)$

l上一点 $(1,1,1)$.

平面: $17(x-1)-9(y-1)-11(z-1)=0$

即 $17x-9y-11z+3=0$

第9章　多变量函数的微分学.

一、重极限

1.（1）$$\lim_{(x,y)\to(0,5)}\frac{e^{xy}-1}{x}=\lim_{(x,y)\to(0,5)}\frac{e^{xy}-1}{xy}\lim_{(x,y)\to(0,5)}y$$

$$=1\times\lim_{y\to5}y$$

$$=5$$

（2）取 $y=kx^{\frac{2}{3}}$

$$\lim_{(x,y)\to(0,0)}\frac{x^2-y^3}{x^2+y^3}=\lim_{(x,y)\to(0,0)}\frac{x^2(1-k^3)}{x^2(1+k^3)}=\frac{1-k^3}{1+k^3}$$

对于不同k（不同的路径），极限值不同.

故极限不存在.

（3）$\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$　　$\cos^4\theta+\sin^4\theta=1-\frac{1}{2}\sin^2 2\theta\in[\frac{1}{2},1]$

$$0\le\left|\frac{x^2+y^2}{x^4+y^4}\right|=\frac{1}{r^2(\sin^4\theta+\cos^4\theta)}\le\frac{2}{r^2}\to0\ (r\to+\infty)$$

由夹逼定理，

$$\lim_{(x,y)\to(0,0)}\frac{x^2+y^2}{x^4+y^4}=0.$$

二、连续与可微

1.（1）$$\lim_{(x,y)\to(0,0)}f(x,y)=\lim_{(x,y)\to(0,0)}(a\sqrt{|x|}+x^2+y^2+b)\frac{\sin(xy^2)}{x^2+y^4}$$

$$=\lim_{(x,y)\to(0,0)}(a\sqrt{|x|}+x^2+y^2+b)\frac{xy^2}{x^2+y^4}\lim_{(x,y)\to(0,0)}\frac{\sin(xy^2)}{xy^2}$$

$$=\lim_{(x,y)\to(0,0)}(a\sqrt{|x|}+x^2+y^2+b)\frac{xy^2}{x^2+y^4}$$

取路径 $x=y^2$.

$$0=f(0,0)=\lim_{(x,y)\to(0,0)}f(x,y)=\lim_{\substack{x=y^2\\y\to0}}(a\sqrt{|x|}+x^2+y^2+b)\frac{xy^2}{x^2+y^4}$$

$$=\lim_{y\to0}(a|y|+y^4+y^2+b)\cdot\frac{y^4}{y^4+y^4}$$

$$=\frac{1}{2}b$$

$\Rightarrow b=0.$

当b=0时. $$\left|\frac{xy^2}{x^2+y^4}\right|\le\frac{xy^2}{2\sqrt{x^2\cdot y^4}}=\frac{1}{2}$$

$$(a\sqrt{|x|}+x^2+y^2)\frac{\sin(xy^2)}{x^2+y^4}\le(a\sqrt{|x|}+x^2+y^2)\frac{xy^2}{x^2+y^4}\le\frac{1}{2}(a\sqrt{|x|}+x^2+y^2)\to0\quad((x,y)\to(0,0))$$

$$\Rightarrow\lim_{(x,y)\to(0,0)}(a\sqrt{|x|}+x^2+y^2)\frac{\sin(xy^2)}{x^2+y^4}=0=f(0,0)$$

在原点连续.

（2）　可微必连续 $\Rightarrow b=0$.

$$f'_x(0,0)=\lim_{x\to0}\frac{f(x,0)-f(0,0)}{x}=0$$

$$f'_y(0,0)=\lim_{y\to0}\frac{f(0,y)-f(0,0)}{y}=0$$

f在原点可微 $\Longleftrightarrow 0=\lim\limits_{(x,y)\to(0,0)}\dfrac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\sqrt{x^2+y^2}}$

$$=\lim_{(x,y)\to(0,0)}\frac{(a\sqrt{|x|}+x^2+y^2)\sin(xy^2)}{(x^2+y^4)\sqrt{x^2+y^2}}$$

$$=\lim_{(x,y)\to(0,0)}\frac{(a\sqrt{|x|}+x^2+y^2)xy^2}{(x^2+y^4)\sqrt{x^2+y^2}}$$

$$\left|\sqrt{x^2+y^2}\,\frac{xy^2}{x^2+y^4}\right|\le\frac{1}{2}\sqrt{x^2+y^2}\to0\quad((x,y)\to(0,0))$$

$$\lim_{(x,y)\to(0,0)}\sqrt{x^2+y^2}\,\frac{xy^2}{x^2+y^4}=0.$$

因此，$$\lim_{(x,y)\to(0,0)}\frac{a\sqrt{|x|}\cdot xy^2}{(x^2+y^4)\sqrt{x^2+y^2}}=0.$$

取 $x=ky^2$，

$$\lim_{\substack{x=ky^2\\y\to0}}\frac{\sqrt{|x|}\cdot xy^2}{(x^2+y^4)\sqrt{x^2+y^2}}=\frac{\sqrt{|k|}\cdot k}{k^2+1}$$

极限值 $\lim\limits_{(x,y)\to(0,0)}\dfrac{\sqrt{|x|}\cdot xy^2}{(x^2+y^4)\sqrt{x^2+y^2}}$ 不存在

因此，$a=0$.

$\Rightarrow a=0,\ b=0.$

2.（1）$$\lim_{(x,y)\to(0,0)}f(x,y)=\lim_{(x,y)\to(0,0)}(x+y)^n\sin\frac{1}{\sqrt{x^2+y^2}}=0\Rightarrow n>0.$$

n为正整数，因此 $n\ge1$.

（2）$$f'_x(0,0)=\lim_{x\to0}\frac{f(x,0)-f(0,0)}{x}=\lim_{x\to0}\frac{x^n\sin\frac{1}{|x|}}{x}=\lim_{x\to0}x^{n-1}\sin\frac{1}{|x|}\ 存在$$

$n-1>0$　即 $n\ge2$

（3）$f(x,y)$在$(0,0)$可微 $\Rightarrow$ 偏导数存在 $\Rightarrow n\ge2$

且 $f'_x(0,0)=0$，$f'_y(0,0)=0$

$f(x,y)$在$(0,0)$可微 $\Longleftrightarrow 0=\lim\limits_{(x,y)\to(0,0)}\dfrac{f(x,y)-f(0,0)-f'_x(0,0)x-f'_y(0,0)y}{\sqrt{x^2+y^2}}$

$$=\lim_{(x,y)\to(0,0)}\frac{(x+y)^n\sin\frac{1}{\sqrt{x^2+y^2}}}{\sqrt{x^2+y^2}}$$

$$=\lim_{r\to0^+}r^{n-1}(\cos\theta+\sin\theta)^n\sin\frac{1}{r}$$

$\Rightarrow n-1>0$. 即 $n\ge2$

（4）偏导数存在 $\Rightarrow n\ge2$

$$f'_x(x,y)=\begin{cases}n(x+y)^{n-1}\sin\frac{1}{\sqrt{x^2+y^2}}-\frac{x(x+y)^n}{(x^2+y^2)^{3/2}}\cos\frac{1}{\sqrt{x^2+y^2}}, & x^2+y^2\neq0.\\ 0, & x^2+y^2=0\end{cases}$$

$$0=\lim_{(x,y)\to(0,0)}f'_x(x,y)=\lim_{(x,y)\to(0,0)}-\frac{x(x+y)^n}{(x^2+y^2)^{3/2}}\cos\frac{1}{\sqrt{x^2+y^2}}$$

$\Rightarrow n>2$　即 $n\ge3$.

3. (1) f在(0,0)处连续.

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{r\to0^+} r^n(\cos\theta+\sin\theta)^n \ln r^2$$

$$= \lim_{r\to0^+} 2r^n \ln r \ (\cos\theta+\sin\theta)^n$$

$$\lim_{r\to0^+} r^n \ln r \overset{x=\ln r}{=\!=\!=} \lim_{x\to-\infty} xe^{nx} = -\lim_{x\to+\infty}\frac{x}{e^{nx}} \overset{\text{L'Hospital}}{=\!=\!=} -\lim_{x\to+\infty}\frac{1}{ne^{nx}}$$

当$n>0$时，$\lim_{r\to0^+} r^n\ln r=0$. 则 $\lim_{(x,y)\to(0,0)} f(x,y)=0$

因此，$n\geq0$.

(2) f在(0,0)处可微 ⇒ 偏导数存在

$$f_x'(0,0) = \lim_{x\to0}\frac{f(x,0)-f(0,0)}{x} = \lim_{x\to0}\frac{x^n\ln x^2-0}{x}$$

$$= \lim_{x\to0} 2x^{n-1}\ln|x|$$

$n>1$时，$f_x'(0,0)$存在. 同理，$f_y'(0,0)$存在.

此时. $f_x'(0,0)=f_y'(0,0)=0$.

f在(0,0)处可微 $\Leftrightarrow 0=\lim_{(x,y)\to(0,0)}\frac{f(x,y)-f(0,0)-f_x'(0,0)x-f_y'(0,0)y}{\sqrt{x^2+y^2}}$

$$=\lim_{(x,y)\to(0,0)}\frac{(x+y)^n\ln(x^2+y^2)}{\sqrt{x^2+y^2}}$$

$$=\lim_{r\to0^+} 2r^{n-1}\ln r \ (\cos\theta+\sin\theta)^n$$

因此. $n-1>0 \Rightarrow n\geq2$.

4. 证: $\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)}\sqrt{|xy|}\lim_{(x,y)\to(0,0)}\frac{\sin(x^2+y^2)}{x^2+y^2} = 0 = f(0,0)$

因此，f在(0,0)处连续.

$$f_x'(0,0) = \lim_{x\to0}\frac{f(x,0)-f(0,0)}{x} = 0$$

$$f_y'(0,0) = \lim_{y\to0}\frac{f(0,y)-f(0,0)}{y} = 0$$

$$\lim_{(x,y)\to(0,0)}\frac{f(x,y)-f(0,0)-f_x'(0,0)x-f_y'(0,0)y}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)}\frac{\sqrt{|xy|}}{(x^2+y^2)^{3/2}}\sin(x^2+y^2)$$

$$=\lim_{(x,y)\to(0,0)}\frac{\sqrt{|xy|}}{\sqrt{x^2+y^2}}\lim_{(x,y)\to(0,0)}\frac{\sin(x^2+y^2)}{x^2+y^2}$$

$$=\lim_{(x,y)\to(0,0)}\frac{\sqrt{|xy|}}{\sqrt{x^2+y^2}}$$

取 $y=kx$，$\lim_{\substack{y=kx\\x\to0}}\frac{\sqrt{|xy|}}{\sqrt{x^2+y^2}} = \frac{\sqrt{|k|}}{\sqrt{1+k^2}}$ 极限值不存在.

故. f在(0,0)处不可微.

5. 证: "⇒" f在(0,0)处可微

$$f(x,y) = f(0,0) + f_x'(0,0)x + f_y'(0,0)y + o(\rho) \quad \rho\to0$$

$$\Rightarrow f_x'(0,0) = \lim_{x\to0}\frac{f(x,0)-f(0,0)}{x} = \lim_{x\to0}\frac{|x|\varphi(x,0)}{x}$$

$$\lim_{x\to0^+}\frac{|x|\varphi(x,0)}{x} = \lim_{x\to0^+}\varphi(x,0) = \varphi(0,0)$$

$$\lim_{x\to0^-}\frac{|x|\varphi(x,0)}{x} = -\lim_{x\to0^-}\varphi(x,0) = -\varphi(0,0)$$

$$\Rightarrow \varphi(0,0)=0.$$

"⇐" $\varphi(0,0)=0$.

$$f_x'(0,0) = \lim_{x\to0}\frac{f(x,0)-f(0,0)}{x} = \lim_{x\to0}\frac{|x|}{x}\varphi(x,0) = 0.$$

$$f_y'(0,0) = \lim_{y\to0}\frac{f(0,y)-f(0,0)}{y} = \lim_{y\to0}\frac{|y|}{y}\varphi(0,y) = 0.$$

$$\left|\frac{f(x,y)-f(0,0)-f_x'(0,0)x-f_y'(0,0)y}{\sqrt{x^2+y^2}}\right| = \frac{|x-y||\varphi(x,y)|}{\sqrt{x^2+y^2}}$$

$$\leq\left(\frac{|x|}{\sqrt{x^2+y^2}}+\frac{|y|}{\sqrt{x^2+y^2}}\right)|\varphi(x,y)| \leq 2|\varphi(x,y)|\to0$$

$$((x,y)\to(0,0))$$

故 f在(0,0)处可微.

6. 证: "⇒" f在(0,0)处可微.

$$f(x,y) = f(0,0) + f_x'(0,0)x + f_y'(0,0)y + o(\rho) \quad \rho\to0$$

$$f_x'(0,0) = \lim_{x\to0}\frac{f(x,0)-f(0,0)}{x} = 0$$

$$f_y'(0,0) = \lim_{y\to0}\frac{f(0,y)-f(0,0)}{y} = 0.$$

$$\frac{f(x,y)-f(0,0)-f_x'(0,0)x-f_y'(0,0)y}{\sqrt{x^2+y^2}} = \frac{|xy|^\alpha}{(x^2+y^2)^{3/2}}$$

$$= r^{2\alpha-3}|\cos\theta\sin\theta| \to 0. \quad (r\to0)$$

$$\Rightarrow 2\alpha-3>0 \quad 即\ \alpha>\frac{3}{2}$$

"⇐" $\alpha>\frac{3}{2}$时，$f_x'(0,0)=0$，$f_y'(0,0)=0$.

$$\frac{f(x,y)-f(0,0)-f_x'(0,0)x-f_y'(0,0)y}{\sqrt{x^2+y^2}} = \frac{|xy|^\alpha}{(x^2+y^2)^{3/2}}$$

$$= r^{2\alpha-3}|\cos\theta\sin\theta|^\alpha \to 0 \quad (r\to0)$$

故 f在(0,0)处可微.

三、微分与偏导数.

1. 证: 取 $y=x$, $\Rightarrow$ $\frac{\partial f}{\partial x}(x,x)=0$, $\frac{\partial f}{\partial y}(x,x)=0$

利用微分中值定理.

$\Rightarrow$ $f(x,x)=f(1,1)=0$.

$$|f(2,0)|=|f(2,0)-f(0,0)|\le\int_0^2\left|\frac{\partial f}{\partial x}(x,0)\right|dx\le\int_0^2 x\,dx=2$$

补充题. 二元函数 $f(x+y)\in C^1(\mathbb{R}^2)$, 且 $f'_x(x,y)=f'_y(x,y)$ 恒成立.

求证: (1) 对于满足 $x_1+y_1=x_2+y_2$ 的任意两点 (x_1,y_1) 与 (x_2,y_2),

都有 $f(x_1,y_1)=f(x_2,y_2)$.

(2) 存在一元函数 $g(x)\in C^1(\mathbb{R})$ 使得 $f(x,y)=g(x+y)$ $((x,y)\in\mathbb{R}^2)$

证: (1). $M_1(x_1,y_1)$, $M_2(x_2,y_2)$

由微分中值定理, $\exists$ 直线段 M_1M_2 上一点 $M(\xi,\eta)$, 使得

$f(x_2,y_2)-f(x_1,y_1)=f'_x(M)(x_2-x_1)+f'_y(M)(y_2-y_1)$

$=f'_x(M)(x_2-x_1+y_2-y_1)=0$.

(2) 由(1)的结论, 在任何一条直线 $x+y=C$ $(C\in\mathbb{R})$ 上 二元函数

$f(x,y)$ 是一个常数. 因此, 存在一元函数 $g(x)$, 使得

$f(x,y)=g(x+y)$. $((x,y)\in\mathbb{R}^2)$

由于 $f(x,y)\in C^1(\mathbb{R}^2)$. $\Rightarrow$ $g(x)\in C^1(\mathbb{R})$.

2. 证: 由微分中值定理, $\exists$ 直线段 AB 上一点 $M(\xi,\eta)$, 使得

$f(A)-f(B)=f'_x(M)(x_1-x_2)+f'_y(M)(y_1-y_2)+f'_z(M)(z_1-z_2)$

$=\nabla u\cdot\vec{BA}$

$|f(A)-f(B)|=|\nabla u\cdot\vec{BA}|\le|\nabla u||\vec{BA}|=|\nabla u|\,\rho(A,B)$

$\le M\rho(A,B)$.

四. 方向导数与梯度

1. $u=xyz$

$\nabla u=(yz,zx,xy)\big|_{(1,1,1)}=(1,1,1)$

$\vec{e}=(\frac{1}{3},-\frac{2}{3},\frac{2}{3})$

$$\frac{\partial u}{\partial e}=\lim_{t\to0}\frac{(1+\frac{1}{3}t)(1-\frac{2}{3}t)(1+\frac{2}{3}t)-1}{t}=\frac{1}{3}$$

2. $\vec{r}=(t^3,2t^2,-2t^3)$

$\vec{r}'(t)=(3t^2,4t,-6t^2)$

$\vec{e}=\frac{1}{\sqrt{61}}(-3,-4,6)$

$$\frac{\partial u}{\partial e}=\lim_{m\to0}\frac{(1-\frac{3}{\sqrt{61}}m)(2-\frac{4}{\sqrt{61}}m)(-2+\frac{6}{\sqrt{61}}m)-(-4)}{m}=\frac{32}{\sqrt{61}}$$

3. (1) $\nabla f=\frac{\partial f}{\partial x}\vec{i}+\frac{\partial f}{\partial y}\vec{j}$

$\frac{\partial f}{\partial x}(0,0)=0$, $\frac{\partial f}{\partial y}(0,0)=0$.

$\nabla f(0,0)=(0,0)$

(2) $$\left|\frac{f(x,y)-f(0,0)-\frac{\partial f}{\partial x}(0,0)x-\frac{\partial f}{\partial y}(0,0)y}{\sqrt{x^2+y^2}}\right|=\left|\frac{\varphi(|xy|)}{\sqrt{x^2+y^2}}\right|$$

$$\le\frac{x^2y^2}{\sqrt{x^2+y^2}}=r^3\cos^2\theta\sin^2\theta\to0\quad(r\to0^+)$$

即 $$\lim_{(x,y)\to(0,0)}\frac{f(x,y)-f(0,0)-\frac{\partial f}{\partial x}(0,0)x-\frac{\partial f}{\partial y}(0,0)y}{\sqrt{x^2+y^2}}=0$$

故 $f(x,y)$ 在 $(0,0)$ 处可微.

4. $\nabla f=(2x,2y,-2)\big|_{(1,1,1)}=(2,2,-2)$

$\vec{n}=(\frac{1}{\sqrt3},\frac{1}{\sqrt3},-\frac{1}{\sqrt3})$

$\frac{\partial u}{\partial \vec{n}}\Big|_M=\nabla u\cdot\vec{n}\big|_M=(\frac{1}{\sqrt3},\frac{1}{\sqrt3},\frac{1}{\sqrt3})\cdot(\frac{1}{\sqrt3},\frac{1}{\sqrt3},-\frac{1}{\sqrt3})=\frac{1}{3}$

5. $$\frac{\partial f}{\partial e}(0,0)=\lim_{t\to0}\frac{f(t\cos\theta,t\sin\theta)-f(0,0)}{t}$$

$$=\lim_{t\to0}\frac{t^4\cos^2\theta\sin^2\theta}{|t|^3}\Big/t$$

$$=\lim_{t\to0}\frac{|t|}{t}\cos^2\theta\sin^2\theta$$

$\theta=0,\frac{\pi}{2},\pi,\frac{3\pi}{2}$ 时, $\frac{\partial f}{\partial e}(0,0)=0$.

其他方向上, $\lim\limits_{t\to0^+}=-\lim\limits_{t\to0^-}$ 极限不存在.

方向导数不存在.

$\frac{\partial f}{\partial x}(0,0)=0$, $\frac{\partial f}{\partial y}(0,0)=0$

$$\lim_{(x,y)\to(0,0)}\frac{f(x,y)-f(0,0)-\frac{\partial f}{\partial x}(0,0)x-\frac{\partial f}{\partial y}(0,0)y}{\sqrt{x^2+y^2}}=\lim_{(x,y)\to(0,0)}\frac{x^2y^2}{(x^2+y^2)^2}$$

取 $y=kx$ $\qquad=\frac{k^2}{(1+k^2)^2}$

极限不存在, 故 f 在 $(0,0)$ 处不可微.

6. $\left.\begin{array}{l}\nabla f\perp(2y^3,-x,0)\\ \nabla f\perp(0,3z^5,-2y^3)\end{array}\right\}\Rightarrow\nabla f\parallel y^3(x,2y^3,3z^5)$

$y\ne0$ 时, $\nabla f\parallel(x,2y^3,3z^5)=\frac{1}{2}\nabla(x^2+y^4+z^6)$.

$x\ne0$ 时, $\begin{cases}u=x^2+y^4+z^6\\ v=y\\ w=z\end{cases}$

$g(u,v,w)=f(x(u,v,w),y(u,v,w),z(u,v,w))$

$\frac{\partial g}{\partial v}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial v}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial v}=0$

$\frac{\partial g}{\partial w}=\frac{\partial f}{\partial x}\frac{\partial x}{\partial w}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial w}+\frac{\partial f}{\partial z}\frac{\partial z}{\partial w}=0$.

$\Rightarrow$ $f(x,y,z)=g(u)=g(x^2+y^4+z^6)$

$\Rightarrow$ f 的等值面为 $x^2+y^4+z^6=C$.

五、复合函数的偏导数.

1. $z=f(\varphi(x+y),x)$

$\frac{\partial z}{\partial y}=f_1'\varphi'$

$\frac{\partial^2 z}{\partial x\partial y}=f_{11}''\varphi'^2+f_{12}''\varphi'+f_1'\varphi''$

2. $u-\varphi(u)=\int_{x-y}^{x+y}P(t)dt$

对x求偏导 $(1-\varphi'(u))\frac{\partial u}{\partial x}=P(x+y)-P(x-y)$

$\varphi'(u)\neq 1 \Rightarrow \frac{\partial u}{\partial x}=\frac{P(x+y)-P(x-y)}{1-\varphi'(u)}$

对y求偏导 $(1-\varphi'(u))\frac{\partial u}{\partial y}=P(x+y)+P(x-y)$

$\varphi'(u)\neq 1 \Rightarrow \frac{\partial u}{\partial y}=\frac{P(x+y)+P(x-y)}{1-\varphi'(u)}$

$\frac{\partial z}{\partial x}=f'(u)\frac{\partial u}{\partial x}$, $\frac{\partial z}{\partial y}=f'(u)\frac{\partial u}{\partial y}$

$\Rightarrow \frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}=2f'(u)\frac{P(x+y)}{1-\varphi'(u)}$

3. $\frac{\partial w}{\partial x}=f_1'+f_2'\cdot yz=f_1'+yzf_2'$

$\frac{\partial w}{\partial x\partial z}=f_{11}''+f_{12}''xy+f_{21}''yz+f_{22}''\cdot xy^2z+f_2'\cdot y$

$=f_{11}''+y(x+z)f_{12}''+xy^2zf_{22}''+yf_2'$

4. $u'(x)=f_1'+f_2'\cdot(f_1'+f_2')$

$u'(1)=a+b(a+b)$

$du\big|_{x=1}=(a+ab+b^2)dx$

5. 证: $\frac{\partial u}{\partial x}=\varphi'(x+y)+\varphi'(x-y)+\psi(x+y)-\psi(x-y)$

$\frac{\partial^2 u}{\partial x^2}=\varphi''(x+y)+\varphi''(x-y)+\psi'(x+y)-\psi'(x-y)$

$\frac{\partial u}{\partial y}=\varphi'(x+y)-\varphi'(x-y)+\psi(x+y)+\psi(x-y)$

$\frac{\partial^2 u}{\partial y^2}=\varphi''(x+y)+\varphi''(x-y)+\psi'(x+y)-\psi'(x-y)$.

$\Rightarrow \frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial y^2}$

6. $u=f(x)g(y)$

$f(x)=xe^x$ $g(y)=ye^y$

$\frac{d^pf}{dx^p}=(x+p)e^x$ $\frac{d^qg}{dy^q}=(y+q)e^y$

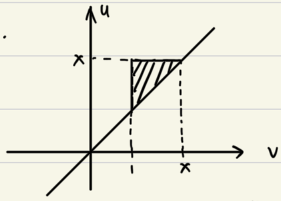
$\Rightarrow \frac{\partial^{p+q}u}{\partial x^p\partial y^q}=(x+p)(y+q)e^{x+y}$

7. 换元 $u=\frac{x}{g(y)}$, $v=y$.

$f(u,v)=ug^2(v)$

$\Rightarrow f(x,y)=xg^2(y)$

$\frac{\partial^2 f}{\partial x\partial y}=2g(y)g'(y)$

8. 

$f(x,y)=x^y+\int_1^x dv\int_v^x e^{-u^2}du$

$=x^y+\int_1^x e^{-u^2}\left(\int_1^u dv\right)du$

$=x^y+\int_1^x (u-1)e^{-u^2}du$

$\frac{\partial f}{\partial x}=yx^{y-1}+(x-1)e^{-x^2}$

$\frac{\partial f}{\partial y}=x^y\ln x$

$\frac{\partial^2 f}{\partial x\partial y}=x^{y-1}+yx^{y-1}\ln x$

9. $\frac{\partial z}{\partial x}=f'\sin y\, e^x$ $\frac{\partial^2 z}{\partial x^2}=f''\sin^2 y\, e^{2x}+f'\sin y\, e^x$

$\frac{\partial z}{\partial y}=f'e^x\cos y$ $\frac{\partial^2 z}{\partial y^2}=f''e^{2x}\cos^2 y-f'e^x\sin y$

$\frac{\partial^2 z}{\partial x^2}+\frac{\partial^2 z}{\partial y^2}=f''e^{2x}=(z+1)e^{2x}$

$\Rightarrow f''=f+1$

10. $\begin{cases} x=\frac{-bu+av}{a-b} \\ y=\frac{u-v}{a-b} \end{cases}$

$\frac{\partial x}{\partial u}=\frac{-b}{a-b}$, $\frac{\partial x}{\partial v}=\frac{a}{a-b}$

$\frac{\partial y}{\partial u}=\frac{1}{a-b}$, $\frac{\partial y}{\partial v}=\frac{-1}{a-b}$

$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\cdot\frac{a}{a-b}+\frac{\partial z}{\partial y}\cdot\frac{-1}{a-b}$

$\frac{\partial^2 z}{\partial u\partial v}=\frac{\partial^2 z}{\partial x^2}\cdot\frac{a}{a-b}\cdot\frac{-b}{a-b}+\frac{\partial^2 z}{\partial y\partial x}\cdot\frac{a}{a-b}\cdot\frac{1}{a-b}$

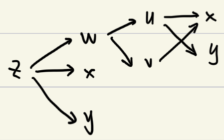
$+\frac{\partial^2 z}{\partial x\partial y}\cdot\frac{-1}{a-b}\cdot\frac{-b}{a-b}+\frac{\partial^2 z}{\partial y^2}\cdot\frac{-1}{a-b}\cdot\frac{1}{a-b}$

$=\frac{1}{(a-b)^2}\left[-ab\frac{\partial^2 z}{\partial x^2}+(a+b)\frac{\partial^2 z}{\partial x\partial y}-\frac{\partial^2 z}{\partial y^2}\right]$

$\Rightarrow \begin{cases} -ab=-\frac{1}{3} \\ a+b=-\frac{4}{3} \end{cases}$

$\Rightarrow \begin{cases} a=-\frac{1}{3} \\ b=-1 \end{cases}$ 或 $\begin{cases} a=-1 \\ b=-\frac{1}{3} \end{cases}$

11.



$$z = \frac{w}{x} + \frac{y}{x} \qquad u = \frac{x}{y},\ v = x$$

$$\frac{\partial z}{\partial y} = \frac{1}{x} + \frac{1}{x}\frac{\partial w}{\partial u}\frac{\partial u}{\partial y} = \frac{1}{x} - \frac{1}{y^2}\frac{\partial w}{\partial u}$$

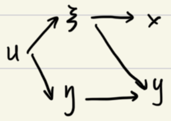
$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{1}{x^2} - \frac{1}{x^2}\frac{\partial w}{\partial u}\frac{\partial u}{\partial y} + \frac{1}{x}\left(\frac{\partial^2 w}{\partial u^2}\frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial v \partial u}\frac{\partial v}{\partial x}\right)\frac{\partial u}{\partial y}$$
$$+ \frac{1}{x}\frac{\partial w}{\partial u}\frac{\partial u}{\partial x\partial y}$$
$$= -\frac{1}{x^2} + \frac{1}{xy^2}\frac{\partial w}{\partial u} - \frac{1}{y^3}\frac{\partial^2 w}{\partial u^2} - \frac{1}{y^2}\frac{\partial^2 w}{\partial v \partial u} - \frac{1}{xy^2}\frac{\partial w}{\partial u}$$
$$= -\frac{1}{x^2} - \frac{1}{y^3}\frac{\partial^2 w}{\partial u^2} - \frac{1}{y^2}\frac{\partial^2 w}{\partial v \partial u}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{1}{x}\frac{\partial^2 w}{\partial u^2}\cdot\left(\frac{\partial u}{\partial y}\right)^2 + \frac{1}{x}\frac{\partial w}{\partial u}\frac{\partial^2 u}{\partial y^2}$$
$$= \frac{x}{y^4}\frac{\partial^2 w}{\partial u^2} + \frac{2}{y^3}\frac{\partial w}{\partial u}$$

$$\Rightarrow y\left(\cancel{\frac{x}{y^4}\frac{\partial^2 w}{\partial u^2}} + \cancel{\frac{2}{y^3}\frac{\partial w}{\partial u}}\right) + x\left(\cancel{-\frac{1}{x^2}} - \cancel{\frac{1}{y^3}\frac{\partial^2 w}{\partial u^2}} - \frac{1}{y^2}\frac{\partial^2 w}{\partial v \partial u}\right)$$
$$+2\left(\cancel{\frac{1}{x}} - \cancel{\frac{1}{y^2}\frac{\partial w}{\partial u}}\right) = \cancel{\frac{1}{x}} - \frac{x}{y}$$

$$\Rightarrow \frac{\partial^2 w}{\partial v \partial u} = \frac{v}{u}$$

11. x, y 自变量.



$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} = -\frac{y}{x^2}\frac{\partial u}{\partial \xi}$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial y} + \frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial y} = \frac{1}{x}\frac{\partial u}{\partial \xi} + \frac{\partial u}{\partial \eta}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{2y}{x^3}\frac{\partial u}{\partial \xi} - \frac{y}{x^2}\frac{\partial^2 u}{\partial \xi^2}\frac{\partial \xi}{\partial x} = \frac{2y}{x^3}\frac{\partial u}{\partial \xi} + \frac{y^2}{x^4}\frac{\partial^2 u}{\partial \xi^2}$$

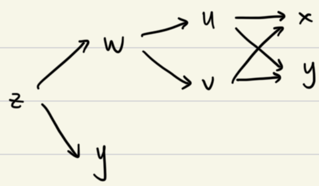
$$\frac{\partial^2 u}{\partial x \partial y} = -\frac{1}{x^2}\frac{\partial u}{\partial \xi} + \frac{1}{x}\frac{\partial^2 u}{\partial \xi^2}\frac{\partial \xi}{\partial x} + \frac{\partial^2 u}{\partial \xi \partial \eta}\frac{\partial \xi}{\partial x}$$
$$= -\frac{1}{x^2}\frac{\partial u}{\partial \xi} - \frac{y}{x^3}\frac{\partial^2 u}{\partial \xi^2} - \frac{y}{x^2}\frac{\partial^2 u}{\partial \xi \partial \eta}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{1}{x}\left(\frac{\partial^2 u}{\partial \xi^2}\frac{\partial \xi}{\partial y} + \frac{\partial^2 u}{\partial \eta \partial \xi}\frac{\partial \eta}{\partial y}\right) + \frac{\partial^2 u}{\partial \xi \partial \eta}\frac{\partial \xi}{\partial y} + \frac{\partial^2 u}{\partial \eta^2}\frac{\partial \eta}{\partial y}$$
$$= \frac{1}{x^2}\frac{\partial^2 u}{\partial \xi^2} + \frac{2}{x}\frac{\partial^2 u}{\partial \eta \partial \xi} + \frac{\partial^2 u}{\partial \eta^2}$$

$$\Rightarrow x^2\left(\cancel{\frac{2y}{x^3}\frac{\partial u}{\partial \xi}} + \cancel{\frac{y^2}{x^4}\frac{\partial^2 u}{\partial \xi^2}}\right) + 2xy\left(-\cancel{\frac{1}{x^2}\frac{\partial u}{\partial \xi}} - \cancel{\frac{y}{x^3}\frac{\partial^2 u}{\partial \xi^2}} - \cancel{\frac{y}{x^2}\frac{\partial^2 u}{\partial \xi \partial \eta}}\right)$$
$$+ y^2\left(\cancel{\frac{1}{x^2}\frac{\partial^2 u}{\partial \xi^2}} + \cancel{\frac{2}{x}\frac{\partial^2 u}{\partial \eta \partial \xi}} + \frac{\partial^2 u}{\partial \eta^2}\right) = 0$$

$$\Rightarrow \frac{\partial^2 u}{\partial \eta^2} = 0.$$

13. x, y 自变量.



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial w}\left(\frac{\partial w}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial x}\right) = \frac{1}{e^y}\left(\frac{1}{2}\frac{\partial w}{\partial u} + \frac{1}{2}\frac{\partial w}{\partial v}\right)$$

$$\frac{\partial z}{\partial y} = -\frac{1}{e^y} + \frac{\partial z}{\partial w}\left(\frac{\partial w}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial w}{\partial v}\frac{\partial v}{\partial y}\right)$$
$$= -\frac{1}{e^y} + \frac{1}{e^y}\left(\frac{1}{2}\frac{\partial w}{\partial u} - \frac{1}{2}\frac{\partial w}{\partial v}\right)$$

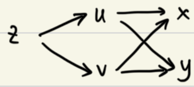
$$\frac{\partial^2 z}{\partial x^2} = \frac{1}{2e^y}\left(\frac{\partial^2 w}{\partial u^2}\frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial v \partial u}\frac{\partial v}{\partial x} + \frac{\partial^2 w}{\partial u \partial v}\cdot\frac{\partial u}{\partial x} + \frac{\partial^2 w}{\partial v^2}\frac{\partial v}{\partial x}\right)$$
$$= \frac{1}{2e^y}\left(\frac{1}{2}\frac{\partial^2 w}{\partial u^2} + \frac{\partial^2 w}{\partial u \partial v} + \frac{1}{2}\frac{\partial^2 w}{\partial v^2}\right)$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-1}{e^y}\left(\frac{1}{2}\frac{\partial w}{\partial u} + \frac{1}{2}\frac{\partial w}{\partial v}\right) + \frac{1}{2e^y}\left(\frac{1}{2}\frac{\partial^2 w}{\partial u^2} - \frac{1}{2}\frac{\partial^2 w}{\partial v^2}\right)$$

$$\Rightarrow \frac{1}{\cancel{2e^y}}\left(\frac{1}{2}\frac{\partial^2 w}{\partial u^2} + \frac{\partial^2 w}{\partial u \partial v} + \cancel{\frac{1}{2}\frac{\partial^2 w}{\partial v^2}}\right) + \cancel{\frac{-1}{e^y}\left(\frac{1}{2}\frac{\partial w}{\partial u} + \frac{1}{2}\frac{\partial w}{\partial v}\right)}$$
$$\frac{1}{\cancel{2e^y}}\left(\frac{1}{2}\frac{\partial^2 w}{\partial u^2} - \cancel{\frac{1}{2}\frac{\partial^2 w}{\partial v^2}}\right) + \cancel{\frac{1}{e^y}\left(\frac{1}{2}\frac{\partial w}{\partial u} + \frac{1}{2}\frac{\partial w}{\partial v}\right)} = 2\frac{w}{\cancel{e^y}}$$

$$\Rightarrow \frac{\partial^2 w}{\partial u^2} + \frac{\partial^2 w}{\partial u \partial v} = 2w$$

14. 证:



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial u^2}\left(\frac{\partial u}{\partial x}\right)^2 + \frac{\partial^2 z}{\partial v \partial u}\frac{\partial u}{\partial x}\frac{\partial v}{\partial x} + \frac{\partial z}{\partial u}\frac{\partial^2 u}{\partial x^2}$$
$$+ \frac{\partial^2 z}{\partial u \partial v}\frac{\partial v}{\partial x}\frac{\partial u}{\partial x} + \frac{\partial^2 z}{\partial v^2}\left(\frac{\partial v}{\partial x}\right)^2 + \frac{\partial z}{\partial v}\frac{\partial^2 v}{\partial x^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial u^2}\left(\frac{\partial u}{\partial y}\right)^2 + \frac{\partial^2 z}{\partial v \partial u}\frac{\partial u}{\partial y}\frac{\partial v}{\partial y} + \frac{\partial z}{\partial u}\frac{\partial^2 u}{\partial y^2}$$
$$\frac{\partial^2 z}{\partial v^2}\left(\frac{\partial v}{\partial y}\right)^2 + \frac{\partial^2 z}{\partial u \partial v}\frac{\partial v}{\partial y}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial^2 v}{\partial y^2}$$

$$\frac{\partial u}{\partial x}\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y}\left(-\frac{\partial u}{\partial y}\right) = -\frac{\partial u}{\partial y}\frac{\partial v}{\partial y}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial x}\right) = -\frac{\partial^2 u}{\partial y^2}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \left(\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2\right)\left(\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}\right)$$

六. 隐函数的偏导数.

1. $3+3z^2\frac{\partial z}{\partial x} = 6yz + 6xy\frac{\partial z}{\partial x} \Rightarrow \frac{\partial z}{\partial x}\Big|_{P_0} = -1$

$$\frac{\partial u}{\partial x} = f'(x^2+y^2+z^2)\left(2x+2z\frac{\partial z}{\partial x}\right)$$

$$\frac{\partial u}{\partial x}\Big|_{P_0} = f'(3)\cdot\left(2+2\times(-1)\right) = 0$$

2. $2x+(2z-4)\frac{\partial z}{\partial x}=0 \Rightarrow \frac{\partial z}{\partial x} = -\frac{x}{z-2}$

$$2y+(2z-4)\frac{\partial z}{\partial y}=0 \Rightarrow \frac{\partial z}{\partial y} = -\frac{y}{z-2}$$

$$dz = -\frac{x}{z-2}dx - \frac{y}{z-2}dy$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{-(z-2)-(-x)\frac{\partial z}{\partial x}}{(z-2)^2} = \frac{-(z-2)^2-x^2}{(z-2)^3}$$

3. $F_x' + F_z'\frac{\partial z}{\partial x} = 0$

$$\Rightarrow \frac{\partial z}{\partial x} = -\frac{F_x'}{F_z'}$$

$$\frac{\partial^2 z}{\partial x} = -\frac{\left(F_{xx}''+F_{xz}''\frac{\partial z}{\partial x}\right)F_z' - F_x'\left(F_{zx}''+F_{zz}''\frac{\partial z}{\partial x}\right)}{F_z'^2}$$

$$= -\frac{F_{xx}''F_z'^2 - 2F_x'F_z'F_{xz}'' + F_x'^2F_{zz}''}{F_z'^3}$$

$$= \frac{-F_{xx}''F_z'^2+2F_x'F_z'F_{xz}''-F_{zz}''F_x'^2}{F_z'^3}$$

4. $\frac{dy}{dx} + f_1'\cdot\left(y+x\frac{dy}{dx}\right) + f_2'\cdot\frac{dz}{dx}=0$

$$\frac{dz}{dx} + g_1'\cdot\left(y+x\frac{dy}{dx}\right) + g_2'\cdot\frac{dz}{dx}=0$$

$$\begin{cases}(1+xf_1')\frac{dy}{dx} + f_2'\frac{dz}{dx} = -yf_1' \\ xg_1'\frac{dy}{dx} + (1+g_2')\frac{dz}{dx} = -yg_1'\end{cases}$$

$$\Rightarrow \frac{dz}{dx} = \frac{xyf_1'g_1' - yg_1'(1+xf_1')}{(1+xf_1')(1+g_2')-xf_2'g_1'} = \frac{-yg_1'}{(1+xf_1')(1+g_2')-xf_2'g_1'}$$

5. $e^{xy}\left(y+x\frac{dy}{dx}\right) - \left(y+x\frac{dy}{dx}\right) = 0 \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$

$xy\neq 0$

$$e^x = \frac{\sin(x-z)}{x-z}\left(1-\frac{dz}{dx}\right) \Rightarrow \frac{dz}{dx} = 1-e^x\frac{x-z}{\sin(x-z)}$$

$$\Rightarrow \frac{du}{dx} = f_1'+f_2'\frac{dy}{dx}+f_3'\frac{dz}{dx}$$

$$= f_1' - \frac{y}{x}f_2' + \left(1-e^x\frac{x-z}{\sin(x-z)}\right)f_3'$$

6. $f_x'+f_y'\varphi'(x)=0$. $f_y'\neq 0$.

$$\varphi'(x) = -\frac{f_x'}{f_y'}$$

$$\varphi''(x) = -\frac{\left(f_{xx}''+f_{xy}''\varphi'(x)\right)f_y' - f_x'\left(f_{yx}''+f_{yy}''\varphi'(x)\right)}{f_y'^2}$$

$$= -\frac{f_{xx}''f_y'^2 - 2f_x'f_y'f_{xy}'' + f_{yy}''f_x'^2}{f_y'^3}$$

7. $F(x,y) = \sin y + e^x - xy - 1 = 0$

$$F_x' = e^x - y \quad F_y' = \cos y - x$$

由隐函数定理.

$$\varphi'(x) = -\frac{F_x'}{F_y'} = -\frac{e^x-y}{\cos y - x}$$

$$\varphi'(0) = -1$$

$$\varphi''(x) = -\frac{\left(e^x-\varphi'(x)\right)(\cos y - x) - (e^x-y)\left(-\sin y\,\varphi'(x)-1\right)}{(\cos y-x)^2}$$

$$\varphi''(0) = -\frac{2\times1-1\times(-1)}{1} = -3$$

8. $3z^2\frac{\partial z}{\partial x} - 3yz - 3xy\frac{\partial z}{\partial x} = 0 \Rightarrow \frac{\partial z}{\partial x} = \frac{yz}{z^2-xy}$

$$3z^2\frac{\partial z}{\partial y} - 3xz - 3xy\frac{\partial z}{\partial y} = 0 \Rightarrow \frac{\partial z}{\partial y} = \frac{xz}{z^2-xy}$$

$$\frac{\partial^2 z}{\partial y\partial x} = \frac{\left(z+y\frac{\partial z}{\partial y}\right)(z^2-xy) - yz\left(2z\frac{\partial z}{\partial y}-x\right)}{(z^2-xy)^2}$$

$$= \frac{\left(z(z^2-xy)+xyz\right)(z^2-xy) - yz\left(2xz^2 - x(z^2-xy)\right)}{(z^2-xy)^3}$$

$z^5 - xyz^3 \quad - xyz^3 - x^2y^2z$

$$= \frac{z(z^4-2xyz^2-x^2y^2)}{(z^2-xy)^3}$$

$$\frac{\partial^2 z}{\partial y\partial x}(0,0,1) = 1$$

9. (1). 证: $\psi(x,y) = g(\varphi(x,y))$

$$\frac{\partial\psi}{\partial x} = g'(\varphi(x,y))\frac{\partial\varphi}{\partial x}$$

$$\frac{\partial\psi}{\partial y} = g'(\varphi(x,y))\frac{\partial\varphi}{\partial y}$$

$$\frac{\partial(\varphi,\psi)}{\partial(x,y)} = \frac{\partial\varphi}{\partial x}\frac{\partial\psi}{\partial y} - \frac{\partial\varphi}{\partial y}\frac{\partial\psi}{\partial x} = 0$$

(2) 证: $\nabla\varphi|_{(0,0)}\neq0$, 不妨设 $\frac{\partial\varphi}{\partial y}\Big|_{(0,0)}\neq0$.

由隐函数定理,

在 $(0,0)$ 附近 存在 $u=\varphi(x,y)$ 确定的

隐函数 $y=y(x,u)$. 则.

$$\frac{\partial y}{\partial x} = -\frac{\varphi_x'}{\varphi_y'}$$

$$\frac{\partial y}{\partial u} = -\frac{1}{\varphi_y'}$$

$$0 = \frac{\partial(\varphi,\psi)}{\partial(x,y)} = \varphi_x'\psi_y' - \psi_x'\varphi_y'$$

$$\psi(x,y) = \psi(x, y(x,u))$$

$$\frac{\partial\psi}{\partial x} = \psi_x' + \psi_y'\frac{\partial y}{\partial x} = \psi_x' - \psi_y'\cdot\frac{\varphi_x'}{\varphi_y'} = 0.$$

故 $\psi(x,y(x,u)) = g(u) = g(\varphi(x,y))$

且 $\frac{dg}{du} = \psi_y'\cdot\frac{\partial y}{\partial u} = \frac{\psi_y'}{\varphi_y'}$ 可微

七、Taylor 展开式.

1. $\sqrt{1-u} = 1 - \frac{1}{2}u - \frac{1}{8}u^2 + o(u^2)$

$f(x,y) = \sqrt{1-(x^2+y^2)} = 1 - \frac{1}{2}(x^2+y^2) - \frac{1}{8}(x^2+y^2)^2 + o(\rho^4)$

2. $z^3 - 2xz + y - 2z + 1 = 0$

$3z^2\frac{\partial z}{\partial x} - 2z - 2x\frac{\partial z}{\partial x} - 2\frac{\partial z}{\partial x} = 0 \Rightarrow \frac{\partial z}{\partial x} = \frac{2z}{3z^2-2x-2}$

$3z^2\frac{\partial z}{\partial y} - 2x\frac{\partial z}{\partial y} + 1 - 2\frac{\partial z}{\partial y} = 0 \Rightarrow \frac{\partial z}{\partial y} = \frac{-1}{3z^2-2x-2}$

$z(0,0) = 1$

$\frac{\partial z}{\partial x}(0,0) = 2 \qquad \frac{\partial z}{\partial y}(0,0) = -1$

$(1+2x-y+ax^2+bxy+cy^2)^3 - 2x(1+2x-y+ax^2+bxy+cy^2) + y$

$-2(1+2x-y+ax^2+bxy+cy^2) + 1 = 0$

$\cancel{1}+\cancel{2x}-\cancel{y}+ax^2+bxy+cy^2+3(4x^2-y^2-4xy)$

$\cancel{-2x}-4x^2+2xy+\cancel{y}+\cancel{1}=0$

$\Rightarrow (a+8)x^2 + (b-10)xy + (c-3)y^2 = 0$

$a=-8,\quad b=10,\quad c=3$

$z = 1+2x-y-8x^2+10xy+3y^2+o(\rho^2)$

八、二元函数的极值.

1. $z = x^4+y^4-x^2-2xy-y^2$

$z'_x = 4x^3-2x-2y=0 \qquad z''_{xx} = 12x^2-2$

$z'_y = 4y^3-2x-2y=0 \qquad z''_{xy} = -2$

$(-1,-1)\ (0,0)\ (1,1) \qquad z''_{yy} = 12y^2-2$

$(-1,-1)$ $A = z''_{xx}(-1,-1) = 10,\ B = z''_{xy} = -2,\ C = z''_{yy}(-1,-1) = 10$

$AC-B^2 = 98>0$ $\quad A>0$ 正定. 极小值

$z(-1,-1) = -2$

$(0,0)$ $A = z''_{xx}(-1,-1) = -2,\ B = z''_{xy} = -2,\ C = z''_{yy}(0,0) = -2$

$AC-B^2 = 0$

在$(0,0)$邻域内, 取$y=x$时, $z=2x^4-4x^2<0=z(0,0)$

取$y=-x$时, $z=2x^4>0=z(0,0)$

$(0,0)$ 不是极值点.

$(1,1)$ $A = z''_{xx}(1,1) = 10,\ B = z''_{xy} = -2,\ C = z''_{yy}(1,1) = 10$

$AC-B^2 = 98>0$ $\quad A>10$ 正定. 极小值

$z(1,1) = -2$

2. 在 $x^2+4y^2\le 4$ 内部.

$$\left.\begin{aligned} z'_x &= 2(2x+3y-6)\cdot 2 = 0 \\ z'_y &= 2(2x+3y-6)\cdot 3 = 0 \end{aligned}\right\} \Rightarrow 2x+3y-6=0.$$

但 $x^2+4y^2 = \left(\frac{6-3y}{2}\right)^2 + 4y^2 = 9-18y+\frac{9}{4}y^2+4y^2$

$= \frac{25}{4}y^2 - 18y + 9$

$= \frac{25}{4}\left(y-\frac{36}{25}\right)^2 + \frac{144}{25} > 4$

即, 直线 $2x+3y-6=0$ 上的点, 都在椭圆外

故在椭圆内部取不到极值.

边界上.

$F = (2x+3y-6)^2 + \lambda(x^2+4y^2-4)$

$$\left\{\begin{aligned} F'_x &= 4(2x+3y-6)+2\lambda x = 0 \\ F'_y &= 6(2x+3y-6)+8\lambda y = 0 \\ & x^2+4y^2-4=0 \end{aligned}\right. \Rightarrow \frac{x}{y} = \frac{8}{3}$$

$$\Rightarrow \begin{cases} x = \frac{8}{5} \\ y = \frac{3}{5} \end{cases} \qquad \begin{cases} x = -\frac{8}{5} \\ y = -\frac{3}{5} \end{cases}$$

$z\left(\frac{8}{5},\frac{3}{5}\right) = 1,\quad z\left(-\frac{8}{5},-\frac{3}{5}\right) = 121$

3. $2x-6y-2y\frac{\partial z}{\partial x} - 2z\frac{\partial z}{\partial x} = 0 \Rightarrow \frac{\partial z}{\partial x} = \frac{x-3y}{y+z} = 0$

$-6x+20y-2z-2y\frac{\partial z}{\partial y} - 2z\frac{\partial z}{\partial y} = 0. \Rightarrow \frac{\partial z}{\partial y} = \frac{-3x+10y-z}{y+z} = 0.$

$x=3y$

$y=z$

$9y^2-18y^2+10y^2-2y^2-y^2+18=0 \Rightarrow y=\pm 3$

$(9,3) \qquad (-9,-3)$

$$\left\{\begin{aligned} &+2-2yz''_{xx}-2z'^2_x-2zz''_{xx}=0 \\ &-6-2z'_x-2yz''_{xy}-2z'_xz'_y-2zz''_{xy}=0 \\ &+20-2z'_y-2z'_y-2yz''_{yy}-2z'^2_y-2zz''_{yy}=0. \end{aligned}\right.$$

$$\left\{\begin{aligned} &+2-2yz''_{xx}-2zz''_{xx}=0 \\ &-6-2yz''_{xy}-2zz''_{xy}=0 \\ &20-2yz''_{yy}-2zz''_{yy}=0 \end{aligned}\right.$$

$(x,y,z) = (9,3,3).$

$A = z''_{xx} = +\frac{1}{6},\quad B = z''_{xy} = -\frac{1}{2} \quad C = z''_{yy} = \frac{5}{3}$

$AC-B^2 = \frac{1}{6}\times\frac{5}{3} - \frac{1}{4} = \frac{1}{36} > 0 \qquad A>0.$

正定. 极小值 $z(9,3) = 3$

$(x,y,z) = (-9,-3,-3)$

$A = z''_{xx} = -\frac{1}{6} \quad B = z''_{xy} = \frac{1}{2} \quad C = z''_{yy} = -\frac{5}{3}$

$AC-B^2 = \frac{1}{36} > 0.$ 负定. 极大值

$z(-9,-3) = -3$

4. $F = x^2+y^2+z^2+\lambda(x^2+y^2-z)+\mu(x+y+z-1)$

$$\begin{cases} F_x' = 2x+2\lambda x+\mu = 0 \\ F_y' = 2y+2\lambda y+\mu = 0 \\ F_z' = 2z-\lambda+\mu = 0 \\ x^2+y^2-z = 0 \\ x+y+z = 1 \end{cases}$$

$2(1+\lambda)(x-y)=0$

若 $\lambda=-1$，则 $\mu=0$，$z=-\frac{1}{2}$，与 $z=x^2+y^2\geq 0$ 矛盾

故 $x=y$.

$$2x^2+2x-1=0$$

$$x=\frac{-1\pm\sqrt{3}}{2}$$

$P_1\ \left(\frac{-1+\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}, 2-\sqrt{3}\right)$ $\quad f(P_1) = 9-5\sqrt{3} = d_{min}^2$

$P_2\ \left(\frac{-1-\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}, 2+\sqrt{3}\right)$ $\quad f(P_2) = 9+5\sqrt{3} = d_{max}^2$

椭圆为有界闭域.

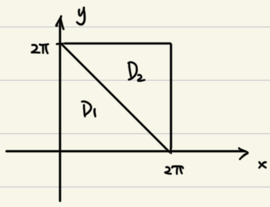
由最值定理，在有界闭域上存在最大、最小值.

$d_{max} = \sqrt{9+5\sqrt{3}}$，$d_{min} = \sqrt{9-5\sqrt{3}}$

5. 令 $g(x,y) = \sin x+\sin y-\sin(x+y)$ $\quad x,y\in\mathbb{R}$

不妨设 $x,y\in[0,2\pi]$

由对称性，取值范围形如 $[-M,M]$. 因此，只要求 M 即可.



$g(2\pi-x, 2\pi-y) = g(x,y)$

只要求 $g(x,y)$ 在 D_1 上的取值范围.

在 D_1 的边界上 $g(x,y)=0$

求 $g(x,y)$ 在 D_1 内驻点 $(x,y>0,\ x+y<2\pi)$

$$\begin{cases} \frac{\partial g}{\partial x} = \cos x-\cos(x+y) = 0 \\ \frac{\partial g}{\partial y} = \cos y-\cos(x+y) = 0. \end{cases}$$

解得 $(x,y) = \left(\frac{2\pi}{3}, \frac{2\pi}{3}\right)$ 唯一驻点.

$g\left(\frac{2\pi}{3}, \frac{2\pi}{3}\right) = \frac{3\sqrt{3}}{2}$

故 $g(x,y)$ 在 D_1 上取值范围为 $\left[0, \frac{3\sqrt{3}}{2}\right]$.

因此，$f(x,y,z)$ 的取值范围为 $\left[-\frac{3\sqrt{3}}{2}, \frac{3\sqrt{3}}{2}\right]$

6. 连续函数 f 在有界闭域 $x^2+y^2+z^2\leq 1$ 上有最大、小值

内部. $f(x,y,z) = x+y-z$

$f_x'=1$，$f_y'=1$，$f_z'=-1$

无驻点，无极值点

边界上，$F = x+y-z+\lambda(x^2+y^2+z^2-1)$

$$\begin{cases} F_x' = 1+2\lambda x = 0 \\ F_y' = 1+2\lambda y = 0 \\ F_z' = -1+2\lambda z = 0 \\ x^2+y^2+z^2 = 1 \end{cases}$$

$\left(-\frac{1}{2\lambda}\right)^2+\left(-\frac{1}{2\lambda}\right)^2+\left(\frac{1}{2\lambda}\right)^2=1 \Rightarrow \frac{3}{4\lambda^2}=1 \Rightarrow \lambda=\pm\frac{\sqrt{3}}{2}$

$M_1\left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$，$\quad f(M_1) = -\sqrt{3}$

$M_2\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$ $\quad f(M_2) = \sqrt{3}$

最大值为 $\sqrt{3}$，最小值为 $-\sqrt{3}$.

7. $F = \frac{x^2+y^2}{2}+\lambda(x^2+y^2+xy-1)$

$$\begin{cases} F_x' = (1+2\lambda)x+\lambda y = 0 \\ F_y' = (1+2\lambda)y+\lambda x = 0 \\ x^2+y^2+xy = 1 \end{cases} \quad y = -\frac{1+2\lambda}{\lambda}x$$

$\lambda=-1$ 时，$x+y=0$. $\quad P_1(1,-1)\ P_2(-1,1)$

$\lambda\neq-1$ 时，$\lambda=-\frac{2}{3}$，$\quad P_3\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)\ P_4\left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$

$z(P_1) = z(P_2) = 1$.

$z(P_3) = z(P_4) = \frac{1}{3}$.

由最值定理，连续函数 z 在有界闭域上有最大、小值

$z_{min} = \frac{1}{3}$，$z_{max} = 1$.

8. 设切点 $P(x_0,y_0,z_0)$ $\quad x_0,y_0,z_0>0$

切平面法向量 $\vec{n} = \nabla F = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, \frac{2z_0}{c^2}\right)$

$$\frac{x_0}{a^2}(x-x_0)+\frac{y_0}{b^2}(y-y_0)+\frac{z_0}{c^2}(z-z_0) = 0$$

即 $\frac{x_0x}{a^2}+\frac{y_0y}{b^2}+\frac{z_0z}{c^2} = 1$

在 x,y,z 轴上截距为 $\frac{a^2}{x_0}$，$\frac{b^2}{y_0}$，$\frac{c^2}{z_0}$.

$$V = \frac{1}{6}\frac{a^2b^2c^2}{x_0y_0z_0}$$

$V(x,y,z) = \frac{a^2b^2c^2}{6xyz}$ $\quad (x,y,z>0)$.

$$F = \frac{a^2b^2c^2}{6xyz}+\lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1\right)$$

$$\begin{cases} F_x' = -\dfrac{a^2b^2c^2}{6x^2yz} + \dfrac{2\lambda x}{a^2} \\ F_y' = -\dfrac{a^2b^2c^2}{6xy^2z} + \dfrac{2\lambda y}{b^2} \\ F_z' = -\dfrac{a^2b^2c^2}{6xyz^2} + \dfrac{2\lambda z}{c^2} \\ \dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}+\dfrac{z^2}{c^2} = 1 \end{cases}$$

$x=\frac{a}{\sqrt3}$, $y=\frac{b}{\sqrt3}$, $z=\frac{c}{\sqrt3}$.

当 $\frac{x}{a}, \frac{y}{b}, \frac{z}{c}$ 中有一个小于 $\frac{1}{3\sqrt3}$ 时.

$$V = \frac{a^2b^2c^2}{6xyz} \geqslant \frac{a^2bc}{6x} > \frac{\sqrt3}{2}abc$$

当 $\frac{x}{a}, \frac{y}{b}, \frac{z}{c} \geq \frac{1}{3\sqrt3}$ 时,

令 $S = \{(x,y,z) \mid \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1, x>0, y>0, z>0\}$.

$S_1 = S \cap \{(x,y,z) \mid \frac{x}{a}, \frac{y}{b}, \frac{z}{c} \geq \frac{1}{3\sqrt3}\}$.

S_1 是有界闭集.

由最值定理, $V(x,y,z)$ 在 S_1 上有最大、最小值.

同时, $\forall (x,y,z) \in S \setminus S_1$, $V > \frac{\sqrt3}{2}abc$

因此, $V(x,y,z)$ 在 S 上的最小值就是在 S_1 上的最小值.

$$V_{min} = V\left(\frac{a}{\sqrt3}, \frac{b}{\sqrt3}, \frac{c}{\sqrt3}\right) = \frac{\sqrt3}{2}abc$$

9. 由最值定理.

连续函数 f 在有界闭域 D 上有最大值、最小值

内部. $f_x' = 2xy(4-x-y) + x^2y(-1) = 8xy - 3x^2y - 2xy^2$

$= xy(8-3x-2y) = 0$

$f_y' = x^2(4-x-y) + x^2y(-1) = x^2(4-x-2y) = 0$.

$$\begin{cases} 3x+2y=8 \\ x+2y=4 \end{cases} \Rightarrow \begin{cases} x=2 \\ y=1 \end{cases}$$

$f(2,1) = 4$

边界上, $x=0$ 上. $f(0,y)=0$

$y=0$ 上, $f(x,0)=0$

$x+y=6$ 上, $f(x,6-x) = -2x^2(6-x) = 2x^2(x-6)$

$4x(x-6) + 2x^2 = 0 \Rightarrow 2x-12+x=0$

$\Rightarrow x=4$

$f(4,2) = -64$

综上所述, $f_{max}=4$, $f_{min}=-64$

10. $\begin{cases} u = x-m \\ v = y-n. \end{cases}$

$5(u+m)^2 - 6(u+m)(v+n) + 5(v+n)^2$

$-6(u+m) + 2(v+n) - 4 = 0$

$5u^2 + 10mu + 5m^2 - 6uv - 6mv - 6nu - 6mn$

$+5v^2 + 10nv + 5n^2 - 6u - 6m + 2v + 2n - 4 = 0$

$5u^2 - 6uv + 5v^2 + (10m-6n-6)u$

$+ (10n-6m+2)v + 5m^2 - 6mn + 5n^2 - 6m + 2n - 4 = 0$

$$\begin{cases} 10m-6n-6=0 \\ 10n-6m+2=0 \end{cases} \Rightarrow m=\frac34,\ n=\frac14$$

$5u^2 - 6uv + 5v^2 - 6 = 0$

连续函数 $f(u,v) = u^2+v^2$ 在有界闭域内有最大、小值.

$F = u^2+v^2 + \lambda(5u^2-6uv+5v^2-6)$

$$\begin{cases} F_u' = 2u + 10\lambda u - 6\lambda v = 0 \\ F_v' = 2v - 6\lambda u + 10\lambda v = 0 \\ 5u^2 - 6uv + 5v^2 - 6 = 0 \end{cases}$$

$(1+5\lambda)u = 3\lambda v$

$16\lambda^2 + 10\lambda + 1 = 0$

$\lambda = -\frac12$ 或 $-\frac18$

$\lambda_1 = -\frac12$, $u=v=\sqrt{\frac32}$ $\quad u^2+v^2=3$

$\lambda_2 = -\frac18$, $u=-v=\sqrt{\frac38}$ $\quad u^2+v^2=\frac34$

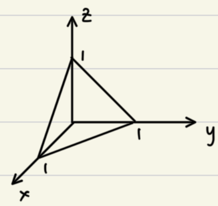
$f_{max}=3$, $f_{min}=\frac34$

$a=\sqrt3$, $b=\frac{\sqrt3}{2}$

$S = \pi ab = \pi\sqrt3 \cdot \frac{\sqrt3}{2} = \frac32\pi$

11. 令 $S = \{(x,y,z) \mid x,y,z \geq 0, x+y+z=1\}$, 有界闭集

f 在 S 上连续. 故 f 在 S 上必有最大值、最小值.



当 x,y,z 中有某个为0时, f 取最小值 0.

所以, 当 f 取最大值时, $x,y,z>0$.

令 $F(x,y,z) = x^a y^b z^c + \lambda(x+y+z-1)$

驻点方程组

$$\begin{cases} F_x' = ax^{a-1}y^bz^c + \lambda = 0 \\ F_y' = bx^ay^{b-1}z^c + \lambda = 0 \\ F_z' = cx^ay^bz^{c-1} + \lambda = 0 \\ x+y+z-1 = 0 \end{cases} \quad (x,y,z>0)$$

$$\Rightarrow \frac{x}{a}=\frac{y}{b}=\frac{z}{c}=\frac{1}{a+b+c}$$

解得. $x=\frac{a}{a+b+c}$, $y=\frac{b}{a+b+c}$, $z=\frac{c}{a+b+c}$

$$f_{max}=f\left(\frac{a}{a+b+c},\frac{b}{a+b+c},\frac{c}{a+b+c}\right)=\frac{a^a b^b c^c}{(a+b+c)^{a+b+c}}$$

注: 初等解法 $x^a y^b z^c \le \frac{a^a b^b c^c}{(a+b+c)^{a+b+c}}$

$$\Leftrightarrow x^{\frac{a}{a+b+c}} y^{\frac{b}{a+b+c}} z^{\frac{c}{a+b+c}} \le \frac{a^{\frac{a}{a+b+c}} b^{\frac{b}{a+b+c}} c^{\frac{c}{a+b+c}}}{a+b+c}$$

令 $p=\frac{a}{a+b+c}$, $q=\frac{b}{a+b+c}$, $r=\frac{c}{a+b+c}$, $p+q+r=1$

$$\Leftrightarrow x^p y^q z^r \le \frac{a^p b^q c^r}{a+b+c}$$

$$\Leftrightarrow x^p y^q z^r \le p^p q^q r^r$$

$$\Leftrightarrow \left(\frac{x}{p}\right)^p \left(\frac{y}{q}\right)^q \left(\frac{z}{r}\right)^r \le 1$$

$$\Leftrightarrow \left(\frac{x}{p}\right)^p \left(\frac{y}{q}\right)^q \left(\frac{z}{r}\right)^r \le p\cdot\frac{x}{p}+q\cdot\frac{y}{q}+r\cdot\frac{z}{r}$$

即为加权平均不等式.

12. (1) $f'_x(x,y)=(y^2+2y)e^x+C(x)$

$f'_x(x,0)=(1+x)e^x \Rightarrow C(x)=(1+x)e^x$

$f'_x(x,y)=(y+1)^2e^x+xe^x$

$f(x,y)=(y+1)^2e^x+xe^x-e^x+C$

$f(0,y)=y^2+2y \Rightarrow y^2+2y+1-1+C=0 \Rightarrow C=0$

$\Rightarrow f(x,y)=(y+1)^2e^x+(x-1)e^x$

$$\begin{cases} f'_x(x,y)=(y+1)^2e^x+xe^x=0 \\ f'_y(x,y)=2(y+1)e^x=0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=-1 \end{cases}$$

$f''_{xx}=(y+1)^2e^x+(x+1)e^x$ $\quad A=f''_{xx}(0,-1)=1$

$f''_{xy}=2(y+1)e^x$ $\quad B=f''_{xy}(0,-1)=0$

$f''_{yy}=2e^x$ $\quad C=f''_{yy}(0,-1)=2$

$\Delta=AC-B^2=2>0$. $A>0$. 正定. 极小值

$f(0,-1)=-1$.

(2) $F=(y+1)^2e^x+(x-1)e^x+\lambda(ye^x-1)$

$$\begin{cases} F'_x=(y+1)^2e^x+xe^x+\lambda ye^x=0 \\ F'_y=2(y+1)e^x+\lambda e^x=0 \\ ye^x=1 \end{cases}$$

消去λ $\Rightarrow y=\sqrt{x+1} \Rightarrow \sqrt{x+1}\,e^x=1$

令 $g(x)=\sqrt{x+1}\,e^x$

$g'(x)=\left(\sqrt{x+1}+\frac{1}{2\sqrt{x+1}}\right)e^x>0 \Rightarrow g(x)\uparrow$

$g(0)=1$.

$\Rightarrow x=0, y=1$. 驻点 $(0,1)$.

$h(x)=f(x,y(x))$

$$h'(x)=f'_x+f'_y\frac{dy}{dx}=f'_x-f'_y\frac{g'_x}{g'_y}=(y+1)^2e^x+xe^x-2(y+1)e^x\frac{ye^x}{e^x}$$
$$=(1-y^2)e^x+xe^x$$

$h''(x)=-2y\frac{dy}{dx}e^x+(1-y^2)e^x+(x+1)e^x=(2+x+y^2)e^x$

$h''(0)=3>0$

$\Rightarrow$ 极小值. $f(0,1)=3$

13、不妨设 $\angle BAC \geqslant \frac{2\pi}{3}$

(1) 当 $|PA|>|AB|+|AC|$ 时, $f(P)>|AB|+|AC|=f(A)$.

所以 f 的最小值在 有界闭集

$|PA|\le|AB|+|AC|$ 上取得.

(2) 设 $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$. $P(x,y)$.

$$f(P)=\sum_{i=1}^{n}\sqrt{(x-x_i)^2+(y-y_i)^2}$$

$$\frac{\partial f}{\partial x}=\sum_{i=1}^{n}\frac{x-x_i}{\sqrt{(x-x_i)^2+(y-y_i)^2}} \quad (P\ne A,B,C)$$

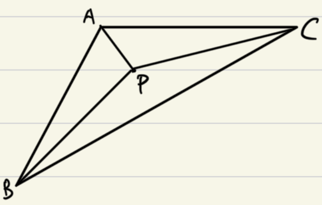
$$\frac{\partial f}{\partial y}=\sum_{i=1}^{n}\frac{y-y_i}{\sqrt{(x-x_i)^2+(y-y_i)^2}}$$

$$\nabla f=-\left(\frac{\vec{PA}}{|\vec{PA}|}+\frac{\vec{PB}}{|\vec{PB}|}+\frac{\vec{PC}}{|\vec{PC}|}\right) \quad (P\ne A,B,C)$$

(3) 假设有驻点, $P(P\ne A,B,C)$, 则.

$$\frac{\vec{PA}}{|\vec{PA}|}+\frac{\vec{PB}}{|\vec{PB}|}+\frac{\vec{PC}}{|\vec{PC}|}=0.$$

则 $\vec{PA},\vec{PB},\vec{PC}$ 两两夹角均为 $\frac{2\pi}{3}$



与$\triangle ABC$有一个内角大于等于 $\frac{2\pi}{3}$ 矛盾.

所以. f的最小值只能在A,B,C三点取到.

$f_{min}=\min\{f(A),f(B),f(C)\}=f(A)=|AB|+|AC|$

九、空间曲线与曲面.

1. $\nabla F = (yz, zx, xy)$

$\nabla G = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2}\right)$ $\quad x,y,z>0.$

$\nabla F \parallel \nabla G$.

即 $\frac{yz}{\frac{2x}{a^2}} = \frac{zx}{\frac{2y}{b^2}} = \frac{xy}{\frac{2z}{c^2}}$

$x:y:z = a:b:c$

$$\begin{cases} x=\frac{a}{\sqrt3} \\ y=\frac{b}{\sqrt3} \\ z=\frac{c}{\sqrt3}\end{cases} \Rightarrow \lambda = \frac{1}{3\sqrt3}abc$$

公共切平面 $\vec n = \left(\frac1a, \frac1b, \frac1c\right)$

公共切平面为 $\frac xa + \frac yb + \frac zc = \sqrt3$

2. $F(x,y,z) = f\left(\frac{y-b}{x-a}, \frac{z-c}{x-a}\right)$

$$\nabla F = \left(f_1' \frac{-(y-b)}{(x-a)^2} + f_2' \frac{-(z-c)}{(x-a)^2},\ f_1'\frac{1}{x-a},\ f_2'\frac{1}{x-a}\right)$$

过曲面上 (x_0,y_0,z_0) 的切平面为

$$\left(f_1' \frac{-(y_0-b)}{(x_0-a)^2} + f_2' \frac{-(z_0-c)}{(x_0-a)^2}\right)(x-x_0) + f_1'\frac{1}{x_0-a}(y-y_0)$$
$$+ f_2'\frac{1}{x_0-a}(z-z_0) = 0.$$

过定点 (a,b,c).

3. $F(x,y,z) = (x^2+y^2) f\left(\frac xy\right) - z^2$

$$\nabla F = \left(2x f\left(\tfrac xy\right) + (x^2+y^2) f'\left(\tfrac xy\right)\frac1y,\ 2y f\left(\tfrac xy\right) + (x^2+y^2) f'\left(\tfrac xy\right)\frac{-x}{y^2},\ -2z\right)$$

过曲面上 (x_0,y_0) 的切平面.

$$G(x,y,z) = \left(2x_0 f\left(\tfrac{x_0}{y_0}\right) + \frac{x_0^2+y_0^2}{y_0} f'\left(\tfrac{x_0}{y_0}\right)\right)(x-x_0) + \left(2y_0 f\left(\tfrac{x_0}{y_0}\right) - \frac{x_0(x_0^2+y_0^2)}{y_0^2} f'\left(\tfrac{x_0}{y_0}\right)\right)(y-y_0)$$
$$-2z_0(z-z_0) = 0$$

$$G(0,0,0) = -2x_0^2 f\left(\tfrac{x_0}{y_0}\right) - \cancel{\frac{x_0(x_0^2+y_0^2)}{y_0} f'\left(\tfrac{x_0}{y_0}\right)} - 2y_0^2 f\left(\tfrac{x_0}{y_0}\right)$$
$$+ \cancel{\frac{x_0(x_0^2+y_0^2)}{y_0} f'\left(\tfrac{x_0}{y_0}\right)} + 2z_0^2$$
$$= -2\left((x_0^2+y_0^2) f\left(\tfrac{x_0}{y_0}\right) - z_0^2\right)$$
$$= 0$$

即，切平面过原点.

4. $P_0(x_0,y_0,z_0)$ 是 $f(x,y,z)$ 在空间光滑曲线 $\Gamma: \begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0.\end{cases}$

上的极值点.

Lagrange 乘数法 $f^* = f + \lambda F + \mu G$

$$\nabla f^* = \nabla f + \lambda \nabla F + \mu \nabla G = \vec 0$$

即 ∇f、∇F、∇G 共面

即 $\nabla f \cdot (\nabla F \times \nabla G) = 0$ $\qquad \nabla f(P_0) \neq \vec 0.\ \nabla F\times\nabla G \neq \vec 0.$

$\nabla f(P_0)$ 为等值面 $f(x,y,z) = f(x_0,y_0,z_0)$ 的法向量 $\vec n$

$\nabla F \times \nabla G$ 为空间光滑曲线 Γ 的切向量. $\vec\tau$

即 $\vec n \perp \vec \tau$

即 等值面 $f(x,y,z) = f(x_0,y_0,z_0)$ 与曲线 Γ 在 P_0 处相切.

5. $F(x,y,z) = x^3 f\left(\frac yx\right) - \sqrt{x^2+y^2+z^2} - z$.

$$\nabla F = \left(3x^2 f\left(\tfrac yx\right) - xy f'\left(\tfrac yx\right) - \frac{x}{\sqrt{x^2+y^2+z^2}},\ x^2 f'\left(\tfrac yx\right) - \frac{y}{\sqrt{x^2+y^2+z^2}},\ -\frac{z}{\sqrt{x^2+y^2+z^2}} - 1\right)$$

切点 (x_0,y_0,z_0)

$$\left(3x_0^2 f\left(\tfrac{y_0}{x_0}\right) - x_0y_0 f'\left(\tfrac{y_0}{x_0}\right) - \frac{x_0}{\sqrt{x_0^2+y_0^2+z_0^2}}\right)(x-x_0) + \left(x_0^2 f'\left(\tfrac{y_0}{x_0}\right) - \frac{y_0}{\sqrt{x_0^2+y_0^2+z_0^2}}\right)(y-y_0)$$
$$+\left(-\frac{z_0}{\sqrt{x_0^2+y_0^2+z_0^2}} - 1\right)(z-z_0) = 0$$

z轴截距为

$$c = z_0 + \frac{\left(3x_0^2 f\left(\tfrac{y_0}{x_0}\right) - \cancel{x_0y_0 f'\left(\tfrac{y_0}{x_0}\right)} - \frac{x_0}{\sqrt{x_0^2+y_0^2+z_0^2}}\right)x_0 + \left(\cancel{x_0^2 f'\left(\tfrac{y_0}{x_0}\right)} - \frac{y_0}{\sqrt{x_0^2+y_0^2+z_0^2}}\right)y_0}{-\frac{z_0}{\sqrt{x_0^2+y_0^2+z_0^2}} - 1}$$

$$= -2\sqrt{x_0^2+y_0^2+z_0^2}$$

$$\Rightarrow \frac{c}{\sqrt{x_0^2+y_0^2+z_0^2}} = -2$$

第10章 多变量函数的重积分

1. $$\iint\limits_{x^2+y^2\le R^2}(3x^2+5y^2)\,dxdy=\int_0^R r^2\cdot r\,dr\int_0^{2\pi}(3\cos^2\theta+5\sin^2\theta)\,d\theta$$
$$=\int_0^R r^3dr\int_0^{2\pi}(5-2\cos^2\theta)\,d\theta$$
$$=\frac{1}{4}R^4\left(4\theta+\sin\theta\cos\theta\right)\Big|_0^{2\pi}$$
$$=2\pi R^4$$

2. $x^2+y^2-2x=0$ 即 $(x-1)^2+y^2=1$.
$$\iint\limits_D xy\,dxdy=\int_0^2 x\left[\int_0^{\sqrt{1-(x-1)^2}}y\,dy\right]dx$$
$$=\int_0^2\frac{1}{2}\left[1-(x-1)^2\right]x\,dx$$
$$=\int_0^2\frac{1}{2}\left(2x-x^2\right)x\,dx$$
$$=\frac{1}{2}\cdot2\cdot\frac{8}{3}-\frac{1}{2}\cdot\frac{16}{4}$$
$$=\frac{2}{3}$$

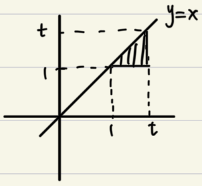
3. $\begin{cases}x=r\cos\theta\\ y=r\sin\theta\end{cases}$

$r^4=a^2r^2\cos2\theta$

$r^2=a^2\cos2\theta$

$r=a\sqrt{\cos2\theta}$ $\qquad 0\le\theta\le\frac{\pi}{4}$
$$\iint\limits_D(x^2+y^2)\,dxdy=4\int_0^{\frac{\pi}{4}}\left[\int_0^{a\sqrt{\cos2\theta}}r^2\cdot r\,dr\right]d\theta$$
$$=4\int_0^{\frac{\pi}{4}}\frac{1}{4}a^4\cos^22\theta\,d\theta$$
$$=a^4\int_0^{\frac{\pi}{4}}\frac{\cos4\theta+1}{2}\,d\theta$$
$$=a^4\left(\frac{1}{8}\sin4\theta+\frac{1}{2}\theta\right)\Big|_0^{\frac{\pi}{4}}$$
$$=\frac{\pi}{8}a^4$$

4. $$F(t)=\int_1^t\left[\int_y^t e^{-x^2}dx\right]dy=\int_1^t e^{-x^2}\left[\int_1^x dy\right]dx$$
$$=\int_1^t(x-1)e^{-x^2}dx$$



$F'(t)=(t-1)e^{-t^2}$

$F'(2)=e^{-4}$

5. $$\int_0^1\left[\int_x^1 e^{y^2}dy\right]dx=\int_0^1 e^{y^2}\left[\int_0^y dx\right]dy$$
$$=\frac{1}{2}e^{y^2}\Big|_0^1$$
$$=\frac{1}{2}(e-1)$$

6. $$\iint\limits_D xy\left[1+x^2+y^2\right]dxdy=\int_0^{2^{\frac14}}\int_0^{\frac{\pi}{2}}r^2\cos\theta\sin\theta\left[1+r^2\right]r\,dr\,d\theta$$
$$=\int_0^{2^{\frac14}}r^3\left[1+r^2\right]dr\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\,d\theta$$
$$=\left(\int_0^1 r^3dr+\int_1^{2^{\frac14}}2r^3dr\right)\frac{1}{2}\sin^2\theta\Big|_0^{\frac{\pi}{2}}$$
$$=\left(\frac{1}{4}+2\times\frac{1}{4}\times(2-1)\right)\times\frac{1}{2}$$
$$=\frac{3}{8}$$

7. $x'(t)=a(1-\cos t)$

$x(t)$ 关于 t 严格 ↗

故. 存在反函数 $t(x)$. 进而 y 是 x 的函数.
$$\iint\limits_D y^2\,dxdy=\int_0^{2\pi a}\left[\int_0^{y(x)}y^2dy\right]dx$$
$$=\int_0^{2\pi}\frac{1}{3}a^3(1-\cos t)^3\cdot a(1-\cos t)\,dt$$
$$=\frac{1}{3}a^4\int_0^{2\pi}(1-\cos t)^4dt$$
$$=\frac{1}{3}a^4\int_0^{\pi}32\sin^8\varphi\,d\varphi$$
$$=\frac{64}{3}a^4\int_0^{\frac{\pi}{2}}\sin^8\varphi\,d\varphi$$
$$=\frac{64}{3}a^4\cdot\frac{\pi}{2}\cdot\frac{7!!}{8!!}$$
$$=\frac{35\pi}{12}a^4$$

8. $$\iint\limits_D xy\,f''_{xy}(x,y)\,dxdy=\int_0^1 x\left[\int_0^1 y\,f''_{xy}(x,y)\,dy\right]dx$$
$$=\int_0^1 x\left[y\,f'_x(x,y)\Big|_0^1-\int_0^1 f'_x(x,y)\,dy\right]dx$$
$$=-\int_0^1\left[x\,f(x,y)\Big|_0^1-\int_0^1 f(x,y)\,dx\right]dy$$
$$=\int_0^1\int_0^1 f(x,y)\,dxdy.$$
$$=\iint\limits_D f(x,y)\,d\sigma$$
$$=a$$

9. 记: $$\int_a^b f(x)\,dx\int_a^b\frac{1}{f(y)}\,dy=\iint\limits_{[a,b]^2}\frac{f(x)}{f(y)}\,dxdy$$
$$=\iint\limits_{[a,b]^2}\frac{f(y)}{f(x)}\,dxdy$$
$$=\frac{1}{2}\iint\limits_{[a,b]^2}\left(\frac{f(x)}{f(y)}+\frac{f(y)}{f(x)}\right)dxdy$$
$$\ge\iint\limits_{[a,b]^2}dxdy=(b-a)^2$$

10. (1) $\frac{1}{2}r(\theta) \le r \le r(\theta)$

$$\iint_D \frac{dxdy}{x^2+y^2} = \iint_D \frac{1}{r^2}\cdot r\,dr\,d\theta = \int_\alpha^\beta d\theta \int_{\frac{1}{2}r(\theta)}^{r(\theta)} \frac{dr}{r} = (\beta - r)\ln 2$$

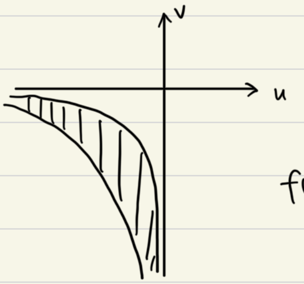
(2) $u = \ln x \quad v = \ln y$

$x = e^u \quad y = e^v$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} e^u & 0 \\ 0 & e^v \end{vmatrix} = e^{u+v}$$

$e^u + e^v = 1$

$e^{2u} + e^{2v} = 1$



$$\begin{cases} u = r\cos\theta \\ v = r\sin\theta \end{cases}$$

$$f(r,\theta) = e^{r\cos\theta} + e^{r\sin\theta} - 1$$

$$\iint_D \frac{dxdy}{xy(\ln^2 x + \ln^2 y)} = \iint_{D'} \frac{du\,dv}{u^2+v^2} = \frac{\pi}{2}\ln 2$$

11. $$f(x,y) = f(0,0) + f_x'(0,0)\,x + f_y'(0,0)\,y + \frac{1}{2}\left(f_{xx}''(0,0)\,x^2 + 2f_{xy}''(0,0)\,xy + f_{yy}''(0,0)\,y^2\right) + o(\rho^2)$$

$$\left|\iint_D f(x,y)\,dxdy\right| = \left|\iint_D \frac{1}{2}\left(f_{xx}''(0,0), \sqrt{2}f_{xy}''(0,0), f_{yy}''(0,0)\right)\cdot(x^2, \sqrt{2}xy, y^2)\,dxdy\right|$$

$$\le \iint_D \frac{1}{2}\left|(f_{xx}'', \sqrt{2}f_{xy}'', f_{yy}'')\cdot(x^2, \sqrt{2}xy, y^2)\right| dxdy$$

$$\le \frac{1}{2}\iint_D \sqrt{\left((f_{xx}'')^2 + 2(f_{xy}'')^2 + (f_{yy}'')^2\right)\cdot(x^4 + 2x^2y^2 + y^4)}\,dxdy$$

$$\le \frac{1}{2}\iint_D (x^2+y^2)\,dxdy$$

$$= \frac{1}{2}\int_0^1 r^3 dr \int_0^{2\pi} d\theta$$

$$= \frac{\pi}{4}$$

12. (1) $$\iint_D g(x,y)\,dxdy = \frac{1}{4}\left[\int_0^1 x(1-x)\,dx\right]^2 = \frac{1}{4}\left[B(2,2)\right]^2$$

$$= \frac{1}{4}\left[\frac{\Gamma(2)\Gamma(2)}{\Gamma(4)}\right]^2$$

$$= \frac{1}{4}\left(\frac{1}{3!}\right)^2$$

$$= \frac{1}{144}$$

(2) 方法一: $\frac{\partial^4 g}{\partial x^2 \partial y^2} = 1$

$$\iint_D f(x,y)\,dxdy = \iint_D f(x,y)\frac{\partial^4 g}{\partial x^2\partial y^2}\,dxdy$$

$$= \int_0^1 \left[\int_0^1 f(x,y)\; d\,\frac{\partial^3 g}{\partial x^2 \partial y}\right] dx$$

$$= \int_0^1 \left[f(x,y)\frac{\partial^3 g}{\partial x^2\partial y}\Big|_{y=0}^{y=1} - \int_0^1 \frac{\partial^3 g}{\partial x^2 \partial y}\frac{\partial f}{\partial y}\,dy\right] dx$$

$$= -\int_0^1 \left[\int_0^1 \frac{\partial f}{\partial y}\; d\,\frac{\partial^2 g}{\partial x^2}\right] dx$$

$$= -\int_0^1 \left[\frac{\partial f}{\partial y}\frac{\partial^2 g}{\partial x^2}\Big|_{y=0}^{y=1} - \int_0^1 \frac{\partial^2 g}{\partial x^2}\frac{\partial^2 f}{\partial y^2}\,dy\right] dx$$

$$= \cdots$$

$$= \int_0^1\int_0^1 g\,\frac{\partial^4 f}{\partial x^2 \partial y^2}\,dxdy$$

$$\Rightarrow \left|\iint_D f(x,y)\,dxdy\right| \le \iint_D |g|\left|\frac{\partial^4 f}{\partial x^2\partial y^2}\right| dxdy \le \iint_D g(x,y)\,dxdy = \frac{1}{144}$$

方法二: 引理. 若 $h(x,y)$ 在 $D = [0,1]\times[0,1]$ 上是 C^4 的.

当 $(x,y)\in\partial D$ 时. $h(x,y) = 0$.

且当 $(x,y)\in D$ 时, $\frac{\partial^4 h}{\partial x^2 \partial y^2} \ge 0$.

则 当 $(x,y)\in D$ 时, $h(x,y) \ge 0$.

引理的证明: 固定 $x_0 \in [0,1]$.

令 $\varphi(y) = \frac{\partial^2 h}{\partial x^2}(x_0, y)$

由条件. $\varphi(0) = 0 = \varphi(1)$ 且 $\varphi''(y) \ge 0$.

故 当 $0 \le y \le 1$ 时, $\varphi(y) \le 0$.

所以 当 $(x,y)\in D$ 时, $\frac{\partial^2 h}{\partial x^2}(x,y) \le 0$.

固定 $y_0 \in [0,1]$, 令 $\psi(x) = h(x, y_0)$.

由条件 $\psi(0) = 0 = \psi(1)$, 且 当 $x\in[0,1]$ 时, $\psi''(x) \le 0$

故 当 $x \in [0,1]$ 时, $\psi(x) \ge 0$.

即 当 $(x,y)\in D$ 时, $h(x,y) \ge 0$.

由引理. 当 $(x,y)\in D$ 时, $g+f \ge 0$, $g-f \le 0$. 即 $|f| \le g$.

因此, $\left|\iint_D f\right| \le \iint_D |f| \le \iint_D g = \frac{1}{144}$